

ALUMNEX CONNECT

Production Deployment & Budget Estimation

Project	Alumnex Connect — Alumni Portal
Technology	MERN Stack + Redis + WebSockets + AI
Document Date	August 2026
Currency	Indian Rupees (₹ INR)

Objective: Publish Alumnex Connect as a market-ready, production-grade SaaS application.

All monetary values in this report are presented in INR. Conversion basis: ₹84 per USD.

Executive Summary

Deployment Stage	Users	Monthly Budget	Yearly Budget
MVP / College Demo	0-500	₹660	₹7,600
Small Production	500-2,000	₹5,880	₹70,560
Medium Production	2,000-10,000	₹26,000	₹3,12,000
Large Scale	10,000-50,000	₹79,800	₹9,57,600

Recommended launch: Start with the MVP configuration at approximately ₹660/month, then move to small production as adoption grows.

1. Project Architecture Overview

Tech Stack Summary

Layer	Technology	Purpose
Frontend	React 18 + Tailwind CSS + Three.js + Framer Motion	SPA with 3D visuals, animations
Backend	Node.js + Express.js	REST API server
Database	MongoDB (Atlas)	Primary data store (23 models)
Cache/PubSub	Redis	Caching, Socket.IO adapter (scaling WebSockets)
Real-time	Socket.IO (with Redis adapter)	Live chat, notifications, WebSocket scaling
File Storage	Cloudinary	Image/file uploads and CDN delivery
Email	Brevo (Sendinblue) SMTP API	OTP verification, notifications, weekly digests
SMS	Fast2SMS	OTP and notification forwarding (India)
AI (Primary)	Groq API (compound-mini)	Career mentor, resume analysis, mock interviews
AI (Fallback)	Google Gemini API (2.0 Flash → 3.x)	Fallback AI engine
Auth	Google OAuth + GitHub OAuth + JWT + OTP	Multi-provider authentication
Email Validation	Disify API	Disposable/invalid email detection
Code Execution	Judge0 CE (RapidAPI)	Coding contest code evaluation (planned)
Dev Stats	GitHub GraphQL + LeetCode + HackerRank + GFG + CodeChef + Codeforces + Duolingo	Developer profile aggregation
Cron Jobs	node-cron	Weekly engagement digest emails
Mobile	Capacitor (Android APK wrapper)	Mobile app wrapper

API Routes (21 Route Modules)

/api/auth → Authentication (Google, GitHub, JWT, OTP, 2FA)
/api/users → User profiles, search, connections
/api/mentorship → Mentorship requests, matching, sessions
/api/jobs → Job listings, applications, AI matching
/api/events → Event creation, RSVP, calendar
/api/forum → Discussion posts, comments, voting
/api/contests → Coding contests, submissions (Judge0)
/api/messages → Real-time chat (Socket.IO), group chats
/api/admin → Admin dashboard, user management
/api/notifications → Push notifications system
/api/upload → File uploads (Cloudinary)
/api/dev-activity → Developer stats aggregation
/api/leaderboard → Gamification leaderboard
/api/feedback → User feedback collection
/api/helpdesk → Support ticket system

/api/projects → Project showcase
/api/institutions → College/institution management
/api/tasks → Task management
/api/ai → AI Career Mentor (Chat, Resume, Interview, Job Match)
/api/tech-hub → Tech news/resources
/api/business → Business directory

MongoDB Models (23 Models)

User, Message, ChatGroup, Contest, DevProfile, Event, Feedback, ForumPost, HelpDesk, Job, JobApplication, JobOpportunity, MentorReview, MentorReward, Mentorship, MentorshipRequest, MentorshipSession, Notification, Project, Report, Task, TechHub, Business

2. Complete Infrastructure Requirements

For production deployment, you need the following infrastructure components:

Component	Minimum Requirement	Recommended
Backend Server	1 vCPU, 1 GB RAM	2 vCPU, 2 GB RAM
MongoDB	512 MB (Atlas M0 Free)	M10 Dedicated (2 GB RAM)
Redis	256 MB	1 GB (for Socket.IO + caching)
Frontend Hosting	Static site CDN	Static site CDN
Domain	1x custom domain	1x `.com` or `.in` domain
SSL Certificate	Required (HTTPS)	Free with most hosts
CI/CD Pipeline	GitHub Actions (free)	GitHub Actions
Monitoring	Basic health checks	APM + error tracking

3. API & Third-Party Service Inventory

All External APIs Used by This Project

#	Service	Purpose	Auth Method	Currently Used In
1	Groq API	AI Chat, Resume Analysis, Mock Interview, Job Match	API Key	`utils/aiHelper.js`, `routes/ai.js`
2	Google Gemini API	AI Fallback Engine	API Key	`utils/aiHelper.js`
3	Brevo (Sendinblue)	Transactional Emails (OTP, Welcome, Digest)	API Key	`utils/sendEmail.js`
4	Fast2SMS	SMS OTP & Notifications (India)	API Key	`utils/sendSMS.js`
5	Cloudinary	Image/File Upload & CDN	API Key + Secret	`config/cloudinary.js`, `routes/upload.js`
6	Google OAuth 2.0	Google Sign-In	Client ID	`services/authService.js`
7	GitHub OAuth	GitHub Sign-In	Client ID + Secret	`services/authService.js`
8	GitHub GraphQL API	Developer Stats (repos, contributions, badges)	Personal Token	`utils/devStatsFetcher.js`
9	LeetCode GraphQL	Coding stats, badges, contest ratings	Public (no key)	`utils/devStatsFetcher.js`
10	HackerRank REST API	Profile, badges, certificates	Public (no key)	`utils/devStatsFetcher.js`
11	GeeksforGeeks Scraper	Coding score, problems solved	Web Scraping	`utils/devStatsFetcher.js`
12	CodeChef Scraper	Rating, stars, division	Web Scraping	`utils/devStatsFetcher.js`

#	Service	Purpose	Auth Method	Currently Used In
13	Codeforces API	Rating, contests, problems solved	Public (no key)	`utils/devStatsFetcher.js`
14	Duolingo API	Language learning stats	Public (no key)	`utils/devStatsFetcher.js`
15	Disify API	Email validation (disposable detection)	Public (no key)	`services/authService.js`
16	Judge0 CE (RapidAPI)	Code execution for contests	API Key	`routes/contests.js` (planned/TODO)
17	Google Userinfo API	Get user profile from OAuth token	Bearer Token	`services/authService.js`
18	GitHub User/Emails API	Get email during OAuth flow	Bearer Token	`services/authService.js`

4. Cloud Hosting Provider Comparison

Backend Server (Node.js + Express + Socket.IO)

Requirement: 1-2 vCPU, 1-2 GB RAM, always-on, WebSocket support, auto-deploy from GitHub

Feature	● Render	● AWS (EC2/ECS)	● Azure (App Service)	● DigitalOcean	● Railway
Starter Plan	₹588/mo (Starter)	₹1,260-18/mo (t3.small)	₹1,092/mo (B1 Basic)	₹504/mo (Basic Droplet)	₹420/mo (Starter)
Production Plan	₹2,100/mo (Standard)	₹1,512/mo (t3.small)	₹1,092/mo (B1)	₹1,008/mo (Regular)	₹1,680/mo (Pro)
WebSocket Support	✓ Native	✓ With ALB	✓ With config	✓ Native	✓ Native
Auto-Deploy (GitHub)	✓ Built-in	△ Needs CodePipeline	△ Needs DevOps	✓ App Platform	✓ Built-in
Free SSL	✓	△ ACM + ALB	✓	✓	✓
Custom Domain	✓	✓	✓	✓	✓
Scaling	Manual/Auto (₹2,100+ Pro)	Auto Scaling Groups	Scale-up/out	Manual	Auto
Cold Start	△ Free tier sleeps	× None	× None	× None	△ Starter sleeps
Hidden Costs	None	EBS, data transfer, ALB (₹1,680+/mo)	Data egress	Bandwidth overages	Bandwidth
Workspace Plan	₹0 (Hobby) / ₹2,100 (Pro)	N/A	N/A	N/A	N/A
Ease of Setup	★★★★★	★★	★★★	★★★★	★★★★★
Best For	Small-Medium apps	Enterprise / custom infra	Microsoft ecosystem	Budget production	Quick deploy

Frontend Hosting (React Static Build)

Provider	Plan	Cost	CDN	Custom Domain	Notes
Render (Static Site)	Free	₹0/mo	✓ Global CDN	✓	Current setup, free forever
Vercel	Hobby	₹0/mo	✓ Edge Network	✓	Best-in-class for React
Netlify	Free	₹0/mo	✓ CDN	✓	Great CI/CD
Cloudflare Pages	Free	₹0/mo	✓ Fastest CDN	✓	Unlimited bandwidth

Provider	Plan	Cost	CDN	Custom Domain	Notes
AWS CloudFront + S3	Pay-as-you-go	₹84-3/mo	✓ CloudFront	✓	Complex setup
Azure Static Web Apps	Free	₹0/mo	✓ Azure CDN	✓	Free for 2 apps

Recommendation: Use Cloudflare Pages (free, fastest CDN, unlimited bandwidth) or Vercel (best React DX).

5. Database Tier Comparison

MongoDB Hosting Options

Provider	Free Tier	Production Tier	Cost/mo	Storage	RAM	Notes
MongoDB Atlas M0	✓ 512 MB	× Not for prod	₹0	512 MB	Shared	Current setup. 100 connections max
MongoDB Atlas M10	×	✓ Dedicated	₹4,788/mo	10 GB	2 GB	Recommended starter production
MongoDB Atlas M20	×	✓ Dedicated	₹12,264/mo	20 GB	4 GB	Medium scale
AWS DocumentDB	×	✓	₹16,800+/mo	Pay per GB	Varies	MongoDB-compatible, expensive
Azure Cosmos DB (MongoDB)	✓ 1000 RU/s	✓	₹2,100+/mo	Pay per GB	Varies	Complex pricing (RU model)
DigitalOcean Managed MongoDB	×	✓	₹1,260/mo	10 GB	1 GB	Simple, affordable
Self-hosted (on VM)	×	✓	₹0 extra	Disk-based	VM RAM	Full control, you manage it

Cost Estimation for Alumnex Connect (23 models, estimated data growth)

Scale	Users	Estimated DB Size	Recommended Tier	Monthly Cost
MVP / College Demo	0 - 500	< 200 MB	Atlas M0 (Free)	₹0
Single College	500 - 2,000	200 MB - 1 GB	Atlas M0 or Shared Cluster	₹0 - ₹756
Multi-College	2,000 - 10,000	1 - 5 GB	Atlas M10 (Dedicated)	₹4,788
Regional Scale	10,000 - 50,000	5 - 20 GB	Atlas M20	₹12,264
National Scale	50,000+	20+ GB	Atlas M30+	₹25,200+

6. Redis/Caching Tier Comparison

Your project uses Redis for: API response caching (30-min TTL), Socket.IO adapter (multi-instance WebSocket scaling)

Provider	Free Tier	Production Tier	Cost/mo	Memory	Notes
Redis Cloud	✓ 30 MB	Essentials	₹0 → ₹420+/mo	30 MB → 250 MB	Official Redis hosting
Render Key Value	✓ 25 MB	Starter	₹0 → ₹840/mo	25 MB → 256 MB	Current provider ecosystem
Upstash Redis	✓ 256 MB, 10K cmd/day	Pay-as-you-go	₹0 → ₹840+/mo	256 MB	Serverless, great free tier
AWS ElastiCache	×	t3.micro	₹1,092/mo	0.5 GB	Enterprise-grade

Provider	Free Tier	Production Tier	Cost/mo	Memory	Notes
Azure Cache for Redis	×	Basic C0	₹1,344/mo	250 MB	Azure ecosystem
Railway Redis	✓ ₹420 credit	Add-on	₹420/mo	Varies	Easy setup

Recommendation: Start with Upstash Redis (256 MB free, serverless, no cold start) for MVP. Move to Redis Cloud Essentials (₹420/mo) for production.

7. API Pricing Deep Dive

7.1 — Groq API (Primary AI Engine)

Used for: AI Career Mentor, Resume Analysis, Mock Interview, Job Matching, Interview Evaluation

Component	Pricing
Free Tier	✓ Available (rate-limited: RPM/TPM limits per org)
Model Tokens (compound-mini)	₹13 / 1M input tokens, ₹50 / 1M output tokens
Web Search (if used)	₹420 - ₹672 / 1,000 requests
Code Execution	₹15 / hour

Estimated Monthly Cost by Scale:

Scale	AI Requests/mo	Estimated Tokens/mo	Monthly Cost
MVP (< 100 users)	~500	~2M tokens	₹0 (Free tier)
Small (500 users)	~2,000	~10M tokens	₹168 - ₹420
Medium (5,000 users)	~15,000	~75M tokens	₹1,260 - ₹3,360
Large (25,000 users)	~75,000	~375M tokens	₹6,300 - ₹16,800

7.2 — Google Gemini API (AI Fallback)

Used as fallback when Groq fails. Model: gemini-2.0-flash (retired June 2026 → migrate to 3.x)

Component	Pricing
Free Tier	✓ Available via Google AI Studio (rate-limited)
Gemini 3.7 Flash	₹63 / 1M input tokens, ₹315 / 1M output tokens
Gemini 3.5 Flash	Lower cost (check current pricing page)

Estimated Cost (fallback only, ~10% of AI traffic):

Scale	Fallback Requests/mo	Monthly Cost
MVP	~50	₹0 (Free tier)
Small	~200	₹0 - ₹84
Medium	~1,500	₹168 - ₹672
Large	~7,500	₹1,260 - ₹4,200

⚠ Action Required: Your code uses `gemini-2.0-flash` which was retired in June 2026. Update `aiHelper.js` to use `gemini-3.5-flash` or `gemini-3.7-flash`.

7.3 — Brevo (Sendinblue) — Transactional Emails

Used for: OTP verification, welcome emails, password resets, peer notifications, weekly Dev Pulse digests

Plan	Cost	Limit	Branding
Free	₹0/mo	300 emails/day (~9,000/mo)	✓ "Sent with Brevo" watermark
Starter	₹756/mo	5,000 emails/mo	△ Watermark (removable for +₹1,008/mo)
Business	₹1,512/mo	5,000 emails/mo	× No watermark
Enterprise	Custom	Unlimited	× No watermark

Estimated Monthly Email Volume:

Scale	OTP Emails	Notifications	Digest	Total/mo	Cost
MVP (100 users)	~200	~300	~400	~900	₹0 (Free)
Small (500 users)	~800	~1,500	~2,000	~4,300	₹0 (Free, ≤300/day)
Medium (5K users)	~3,000	~10,000	~20,000	~33,000	₹2,100/mo (Starter)
Large (25K users)	~10,000	~50,000	~100,000	~160,000	₹5,460/mo (Business)

7.4 — Fast2SMS — SMS Notifications (India Only)

Used for: OTP forwarding, notification SMS (Indian phone numbers only)

Wallet Amount	Per SMS Cost	Notes
₹100 - ₹3,999	₹0.25	+ GST
₹4,000 - ₹7,999	₹0.21	+ GST
₹8,000 - ₹13,999	₹0.19	+ GST
₹14,000+	₹0.17	+ GST
Quick SMS (No DLT)	₹5.00	International route, expensive

Estimated Monthly Cost:

Scale	SMS Sent/mo	Cost (₹0.25/SMS)	Cost (₹0.17/SMS)
MVP	~100	₹25 (₹25)	₹17
Small	~500	₹125 (₹126)	₹85
Medium	~3,000	₹750 (₹756)	₹510
Large	~15,000	₹3,750 (₹3,780)	₹2,550

Note: DLT registration is required for production SMS in India. Register on your telecom provider's DLT portal (Jio, Airtel, Vi, or BSNL). Cost: Free but requires verification.

7.5 — Cloudinary — File Uploads & Image CDN

Used for: Profile photos, event images, project screenshots, resume uploads

Plan	Cost	Credits	Storage	Bandwidth	Transformations
Free	₹0/mo	25 credits	25 GB storage	25 GB bandwidth	25K transforms
Plus	₹7,476/mo	225 credits	225 GB	225 GB	225K transforms
Advanced	₹20,916/mo	1,000 credits	1,000 GB	1,000 GB	1M transforms

Estimated Usage:

Scale	Uploads/mo	Storage (total)	Bandwidth/mo	Plan Needed	Cost
MVP	~200	< 2 GB	< 5 GB	Free	₹0
Small	~1,000	5 - 10 GB	10 - 20 GB	Free	₹0
Medium	~5,000	20 - 50 GB	50 - 100 GB	Plus	₹7,476/mo
Large	~20,000	100+ GB	200+ GB	Advanced	₹20,916/mo

7.6 — Judge0 CE (RapidAPI) — Code Execution

Used for: Coding contest submissions (currently marked as TODO in code)

Plan	Cost	Submissions
Basic (Free)	₹0/mo	~50/day (shared key, unreliable)
Pay-Per-Use	₹0/submission	Unlimited
Self-Hosted (Docker)	VM cost only	Unlimited

Estimated Cost (if implemented):

Scale	Submissions/mo	RapidAPI Cost	Self-Hosted Cost
MVP	~200	₹29	₹420/mo (small VM)
Small	~2,000	₹286	₹840/mo
Medium	~10,000	₹1,428	₹1,260/mo
Large	~50,000	₹7,140	₹2,100/mo

Recommendation: Self-host Judge0 on a dedicated container for cost efficiency at scale.

7.7 — Google OAuth 2.0

Component	Cost
OAuth Usage	FREE (unlimited requests)
Consent Screen Verification	FREE (standard scopes: email, profile)
Security Assessment	NOT REQUIRED (you only use basic profile scopes)

Your app only requests `email` and `profile` scopes — no restricted/sensitive scopes required. Cost is ₹0.

7.8 — GitHub OAuth & GraphQL API

Component	Cost
GitHub OAuth	FREE
GitHub GraphQL API	FREE (5,000 points/hour with PAT)

Component	Cost
GitHub REST API	FREE (5,000 requests/hour with PAT)

⚠ Production Note: Generate a dedicated GitHub PAT (Fine-grained token) for DevPulse stats fetching. The free tier is sufficient for up to ~10K users.

7.9 — Disify (Email Validation)

Plan	Cost	Limit
Anonymous	₹0	1,000 validations/day
Free Account	₹0	10,000 validations/day
Starter	₹756/mo	Unlimited (120 RPM)

Recommendation: Free account is sufficient for up to ~10K registrations/day.

7.10 — Public APIs (No Cost)

API	Rate Limit	Cost
LeetCode GraphQL	Generous (unofficial)	FREE
HackerRank REST	Generous (unofficial)	FREE
Codeforces API	1 request/2 seconds	FREE
CodeChef (Scraping)	Depends on traffic	FREE
GFG (Scraping)	Depends on traffic	FREE
Duolingo API	Generous (unofficial)	FREE

⚠ Risk: LeetCode, HackerRank, GFG, CodeChef, and Duolingo are unofficial/scraped APIs. They may break without notice. Consider adding error handling and caching (already implemented via Redis).

8. Domain, SSL & CDN Costs

Domain Registration

Domain	Namecheap	GoDaddy	Notes
alumnexconnect.com	₹588 (1st yr) → ₹1,512/yr renewal	₹1,932/yr	Recommended
alumnexconnect.in	₹840/yr	~₹899/yr (₹924)	India-specific
alumnex.app	₹1,260/yr	₹1,680/yr	Modern TLD

SSL Certificate

Provider	Cost	Notes
Render	FREE (auto-provisioned Let's Encrypt)	Current setup
Cloudflare	FREE (Edge SSL + Origin CA)	Best option
Let's Encrypt	FREE	Self-managed on VMs
Namecheap PositiveSSL	₹1,092/yr	If self-hosting

CDN (Content Delivery Network)

Provider	Free Tier	Paid	Notes
Cloudflare	✓ Unlimited bandwidth	Pro: ₹1,680/mo	Best free CDN globally
Render CDN	✓ (with Static Site)	Included	Already using
AWS CloudFront	1 TB/mo (12 months)	₹7/GB	Complex setup

Recommendation: Use Cloudflare Free as your DNS + CDN + SSL layer (in front of Render/any provider).

9. Production Optimization Checklist

Critical Items Before Going Live

#	Task	Status	Priority
1	Rotate ALL API keys & secrets (exposed in .env file)	● CRITICAL	P0
2	Set strong JWT_SECRET (currently placeholder value)	● CRITICAL	P0
3	Update Gemini model from `gemini-2.0-flash` to `gemini-3.x`	● Required	P1
4	Remove self-ping hack (Render free tier workaround)	● Production	P2
5	Add error monitoring (Sentry free: 5K events/mo)	● Required	P1
6	Add health check endpoint (`/api/health`)	● Required	P1
7	Enable MongoDB Atlas IP allowlist (remove `0.0.0.0/0`)	● Required	P1
8	Register DLT for production SMS (India telecom requirement)	● Required	P1
9	Google OAuth consent screen verification (for >100 users)	● Required	P1
10	Implement Judge0 integration (currently TODO)	● Nice to have	P2
11	Add Redis connection pooling config	● Optimization	P2
12	Enable gzip/brotli compression (already has `compression` middleware)	✓ Done	—
13	Set up CI/CD pipeline (GitHub Actions)	● Recommended	P2
14	Add rate limiting for AI endpoints (prevent abuse)	● Required	P1
15	Configure production logging (structured JSON logs)	● Recommended	P2

SECURITY ALERT: Your `.env` file currently contains real API keys, database credentials, and OAuth secrets. Before deploying to production:

1. Rotate ALL keys immediately (MongoDB password, Brevo key, Cloudinary key, GitHub secrets, Groq key, Fast2SMS key)

2. Never commit `.env`` files to Git (verify `.gitignore`` includes it)

3. Use environment variables in your hosting provider's dashboard instead

10. Budget Summary Tables

10.1 — MVP / College Demo Budget (0-500 users)

Goal: Deploy a working version for a single college with minimal cost.

Service	Provider	Plan	Monthly Cost (INR)
Backend Server	Render	Starter (₹588)	₹588
Frontend Hosting	Render / Cloudflare Pages	Free	₹0
Database	MongoDB Atlas	M0 Free	₹0
Redis/Cache	Upstash / Redis Cloud	Free	₹0
Email (Brevo)	Brevo	Free (300/day)	₹0
SMS (Fast2SMS)	Fast2SMS	Wallet (~100 SMS)	~₹25
AI (Groq)	Groq	Free Tier	₹0
AI Fallback (Gemini)	Google	Free Tier	₹0
File Uploads	Cloudinary	Free (25 credits)	₹0
Email Validation	Disify	Free	₹0
Domain	Namecheap	`.com` 1st year	~₹50
SSL	Render / Cloudflare	Free	₹0
Monitoring	Sentry	Free (5K events/mo)	₹0
CDN	Cloudflare	Free	₹0
		TOTAL (Monthly)	~₹660
		TOTAL (Yearly)	~₹7,600

10.2 — Small Production Budget (500-2,000 users)

Goal: Reliable production deployment for 1-3 colleges.

Service	Provider	Plan	Monthly Cost (INR)
Backend Server	Render	Standard (₹2,100)	₹2,100
Workspace Plan	Render	Pro (₹2,100)	₹2,100
Frontend	Cloudflare Pages	Free	₹0
Database	MongoDB Atlas	Shared Cluster	₹756
Redis	Upstash	Pay-as-you-go	₹420
Email (Brevo)	Brevo	Free (300/day)	₹0
SMS	Fast2SMS	~500 SMS/mo	~₹125
AI (Groq)	Groq	Pay-as-you-go	₹252
File Uploads	Cloudinary	Free	₹0
Domain	Namecheap	`.com` renewal	~₹126

Service	Provider	Plan	Monthly Cost (INR)
CDN + SSL	Cloudflare	Free	₹0
Monitoring	Sentry	Free	₹0
		TOTAL (Monthly)	~₹5,880
		TOTAL (Yearly)	~₹70,560

10.3 — Medium Production Budget (2,000-10,000 users)

Goal: Multi-college deployment with dedicated resources.

Service	Provider	Plan	Monthly Cost (INR)
Backend Server	Render Standard / AWS t3.small	Production	₹2,100
Workspace/Infra	Render Pro / AWS overhead	—	₹2,100
Frontend	Cloudflare Pages	Free	₹0
Database	MongoDB Atlas	M10 Dedicated	₹4,788
Redis	Redis Cloud Essentials	250 MB	₹420
Email (Brevo)	Brevo	Starter (20K/mo)	₹2,100
SMS	Fast2SMS	~3K SMS/mo	~₹750
AI (Groq)	Groq	Pay-as-you-go	₹2,100
AI (Gemini)	Google	Pay-as-you-go	₹420
File Uploads	Cloudinary	Plus (₹7,476)	₹7,476
Judge0	RapidAPI	Pay-per-use	₹1,428
Domain	Namecheap	`.com`	~₹126
CDN + SSL	Cloudflare	Free	₹0
Monitoring	Sentry	Team (₹2,184/mo)	₹2,184
		TOTAL (Monthly)	~₹26,000
		TOTAL (Yearly)	~₹3,12,000

10.4 — Large Scale Budget (10,000-50,000 users)

Goal: National-level platform with high availability.

Service	Provider	Plan	Monthly Cost (INR)
Backend Server	AWS ECS Fargate / Azure	2x instances	₹5,040
Load Balancer	AWS ALB / Azure LB	—	₹1,680
Frontend	Cloudflare Pages	Pro	₹1,680
Database	MongoDB Atlas	M20 Dedicated	₹12,264
Redis	Redis Cloud Essentials	1 GB	₹2,688
Email (Brevo)	Brevo	Business (60K/mo)	₹5,460
SMS	Fast2SMS	~15K SMS/mo	~₹3,750
AI (Groq)	Groq	Pay-as-you-go	₹12,600
AI (Gemini)	Google	Pay-as-you-go	₹2,940

Service	Provider	Plan	Monthly Cost (INR)
File Uploads	Cloudinary	Advanced	₹20,916
Judge0	Self-hosted (VM)	Dedicated	₹2,100
Domain	—	`.com`	~₹126
CDN + SSL	Cloudflare	Pro	₹1,680
Monitoring	Sentry + Datadog	Team	₹4,200
Backup/DR	Atlas + S3	—	₹2,520
		TOTAL (Monthly)	~₹79,800
		TOTAL (Yearly)	~₹9,57,600

10.5 — Provider Comparison (Total Monthly Cost for Medium Scale)

Stack	Backend	DB	Redis	Email	CDN	Total/mo
● Render Stack	₹4,200	₹4,788 (Atlas)	₹840	₹2,100	₹0	₹23,520
● AWS Stack	₹3,192 (t3.small + ALB)	₹4,788 (Atlas)	₹1,092 (ElastiCache)	₹2,100	₹252	₹26,880
● Azure Stack	₹1,092 (B1)	₹2,100 (Cosmos DB)	₹1,344	₹2,100	₹0	₹22,260
● DigitalOcean Stack	₹1,008	₹1,260 (DO MongoDB)	₹840	₹2,100	₹0	₹16,800
● Railway Stack	₹1,680	₹4,788 (Atlas)	₹420	₹2,100	₹0	₹20,580

Winner for Cost: DigitalOcean (₹16,800/mo)

Winner for Ease: Render or Railway (₹20,580-280/mo)

Winner for Scale: AWS (₹26,880/mo but infinite scaling)

11. Scaling Roadmap & Cost Projections

Phase	Timeline	Users	Monthly Budget	Key Upgrades
Phase 1: MVP	Month 1-3	0-500	₹672/mo (₹660)	Free tiers everywhere
Phase 2: Validation	Month 3-6	500-2K	₹5,880/mo (₹5,880)	Paid backend, shared DB
Phase 3: Growth	Month 6-12	2K-10K	₹26,040/mo (₹26,000)	Dedicated DB, paid email, AI
Phase 4: Scale	Year 2+	10K-50K	₹79,800/mo (₹79,800)	Multi-instance, load balancer

12. Recommended Production Stack

Best Value Stack for Launch (Recommended)

Component	Provider	Plan	Cost/mo
Backend	Render	Starter (₹588) → Standard (₹2,100)	₹588 - ₹2,100
Frontend	Cloudflare Pages	Free	₹0

Component	Provider	Plan	Cost/mo
Database	MongoDB Atlas	M0 Free → M10 (₹4,788)	₹0 - ₹4,788
Redis	Upstash	Free → Pay-as-you-go	₹0 - ₹420
Email	Brevo	Free (300/day)	₹0
SMS	Fast2SMS	Wallet	₹84
AI	Groq	Free tier → Pay-as-you-go	₹0 - ₹420
CDN + DNS + SSL	Cloudflare	Free	₹0
Domain	Namecheap	`.com`	₹84/mo
Monitoring	Sentry	Free	₹0
TOTAL		MVP Launch	₹672 - ₹7,896/mo

Immediate Action Items to Go Live

1. Buy domain → `alumnexconnect.com` on Namecheap (₹588)
2. Set up Cloudflare → Point domain, enable free CDN + SSL
3. Deploy backend → Render Starter (₹588/mo) with all env vars
4. Deploy frontend → Cloudflare Pages (free) with `REACT\_APP\_API\_URL`
5. Rotate all keys → Generate fresh secrets for all APIs
6. Google OAuth verification → Submit consent screen for >100 users
7. DLT registration → Register for production SMS (if needed)
8. Update Gemini model → Change from `gemini-2.0-flash` to `gemini-3.7-flash`
9. Test everything → Run through all features on production
10. Set up monitoring → Free Sentry account for error tracking

Bottom Line: You can launch Alumnex Connect as a production SaaS for as low as ₹660/month (₹672/mo) using free tiers strategically. As you grow to 10,000+ users, expect costs to reach ₹26,000/month (₹26,040/mo). The platform is architecturally sound for scaling — the main cost drivers at scale are Cloudinary (file storage), MongoDB Atlas (database), and AI APIs (Groq/Gemini).

This document was prepared based on analysis of the Alumnex Connect codebase and current market pricing as of August 2026. Prices are subject to change — always verify on the provider's official pricing page before committing.